Error Budgeting & Imaging using Macro-Scale AFM

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Overview

This is a final project for ME392: Advanced Mechatronic II. Using a tapping mode macro-AFM which can track a sample in a vertical (z) position and moving stage in a horizontal (x-y) plane, an object can be sampled to acquire an image. This project involves error modeling for the imaging stages and experiments that aims to scan over a nickel coin using LabVIEW, a programmable moving stage, and macro-AFM.

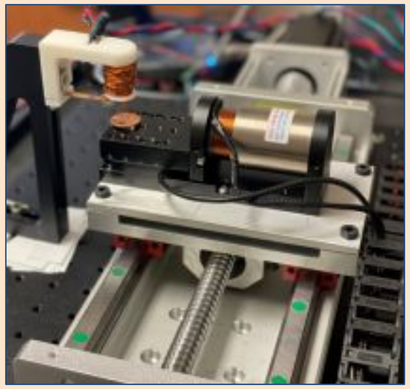


Figure 1: Experiment Setup



Figure 2: Sample for Scanning

Dynamic Error Budgeting

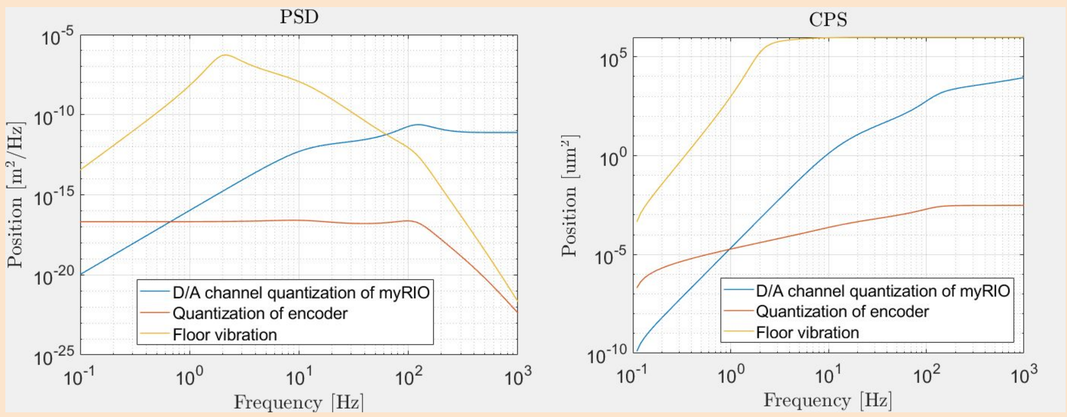


Figure 3: Output Plots of PSD and CPS

Imaging

Image Parameters

- Scanned Area: 10 mm x 10 mm
- x line of rate: 100 s / line
- y line of rate: 100 s / line
- MATLAB colormap: default (left)
- Matplotlib colormap: coolwarm (right)

AFM Parameters

- Reference Amplitude: 0.08
- Sampling Frequency: 2000
- Resonance Frequency: 43 Hz
- Locked Frequency:44 Hz
- Ti Tracking: 4
- Kp: 0.64

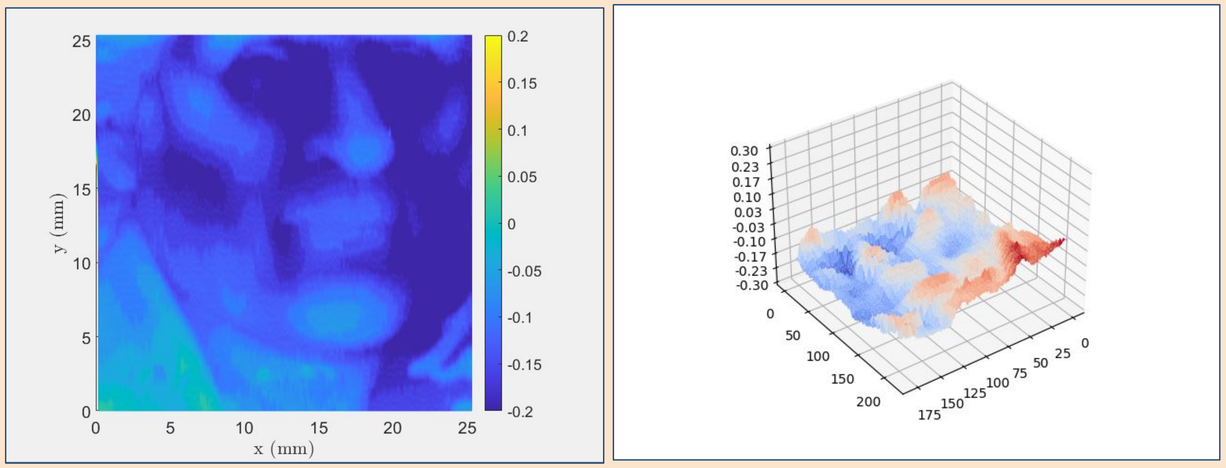


Figure 4: Colormap of the Result: MATLAB (left) Python (right)